

WEEK 3 – FINANCIAL DATA IMPORT, TRANSFORMATION & TIME INTELLIGENCE

Shopify Stock Data Analysis using Power BI

INTRODUCTION

This project focuses on importing and preparing time-series Shopify stock market data using Power BI. A dedicated Date Table was created for time-based analysis, financial DAX measures were developed, and interactive visualizations were used to analyze stock performance.

The analysis covers **Open Price, High Price, Low Price, Close Price, Adjusted Close, and Trading Volume** over time.

Figure 1: Project requirements and objectives for Shopify Stock Data.

OBJECTIVES

The main objectives of this project are:

- To import and prepare time-series stock market data using Power Query.
- To create a dedicated Date Table for time-based analysis.
- To establish a date-based relationship between the stock data and Date Table.
- To create DAX measures for Open Price, Close Price, High Price, Low Price, Trading Volume, and return-related analysis.
- To perform time-intelligence calculations such as YTD and prior-period analysis.
- To create a Year–Quarter–Month–Day hierarchy for drill-down analysis.

- To build KPI cards, charts, slicers, and a Decomposition Tree.
- To create a structured Power BI report for financial analysis.

DATASET DESCRIPTION

The **Shopify Stock Data** dataset contains daily time-series stock information.

The main fields are:

- Date
- Open
- High
- Low
- Close
- Adjusted Close
- Volume

The provided CSV file contains **2,469 records and 7 columns**.

STEP 1 – IMPORT AND PREPARE THE DATA

The Shopify stock dataset was imported into Power BI. The date field was prepared for time-based analysis, while the stock price and trading-volume fields were maintained as numeric fields.

The following data preparation activities were performed:

- Checked stock data fields and data types.
- Prepared the Date field for time-based analysis.
- Kept price and volume fields as numeric analysis fields.
- Created separate Date and Measures tables.

SHOP table, Date Table, and Measures Table in the Power BI model.

Date	Year	Quarter	Month	Month Number	Day
21-05-2015 00:00:00	2015	Q2	May	5	21
22-05-2015 00:00:00	2015	Q2	May	5	22
23-05-2015 00:00:00	2015	Q2	May	5	23
24-05-2015 00:00:00	2015	Q2	May	5	24
25-05-2015 00:00:00	2015	Q2	May	5	25
26-05-2015 00:00:00	2015	Q2	May	5	26
27-05-2015 00:00:00	2015	Q2	May	5	27
28-05-2015 00:00:00	2015	Q2	May	5	28
29-05-2015 00:00:00	2015	Q2	May	5	29
30-05-2015 00:00:00	2015	Q2	May	5	30
31-05-2015 00:00:00	2015	Q2	May	5	31
01-06-2015 00:00:00	2015	Q2	June	6	1
02-06-2015 00:00:00	2015	Q2	June	6	2
03-06-2015 00:00:00	2015	Q2	June	6	3
04-06-2015 00:00:00	2015	Q2	June	6	4
05-06-2015 00:00:00	2015	Q2	June	6	5
06-06-2015 00:00:00	2015	Q2	June	6	6
07-06-2015 00:00:00	2015	Q2	June	6	7
08-06-2015 00:00:00	2015	Q2	June	6	8
09-06-2015 00:00:00	2015	Q2	June	6	9
10-06-2015 00:00:00	2015	Q2	June	6	10
11-06-2015 00:00:00	2015	Q2	June	6	11
12-06-2015 00:00:00	2015	Q2	June	6	12
13-06-2015 00:00:00	2015	Q2	June	6	13
14-06-2015 00:00:00	2015	Q2	June	6	14
15-06-2015 00:00:00	2015	Q2	June	6	15
16-06-2015 00:00:00	2015	Q2	June	6	16
17-06-2015 00:00:00	2015	Q2	June	6	17
18-06-2015 00:00:00	2015	Q2	June	6	18
19-06-2015 00:00:00	2015	Q2	June	6	19
20-06-2015 00:00:00	2015	Q2	June	6	20
21-06-2015 00:00:00	2015	Q2	June	6	21
22-06-2015 00:00:00	2015	Q2	June	6	22

STEP 2 – CREATE THE DATE TABLE

A separate **Date Table** was created using DAX. It provides a continuous date range based on the minimum and maximum dates available in the SHOP table.

Date Table =CALENDAR(MIN('SHOP'[date]), MAX('SHOP'[date]))

The Date Table contains the following fields:

- Date
- Year
- Quarter
- Month
- Month Number
- Day

Date Table fields in the completed model.

STEP 3 – ESTABLISH TABLE RELATIONSHIP

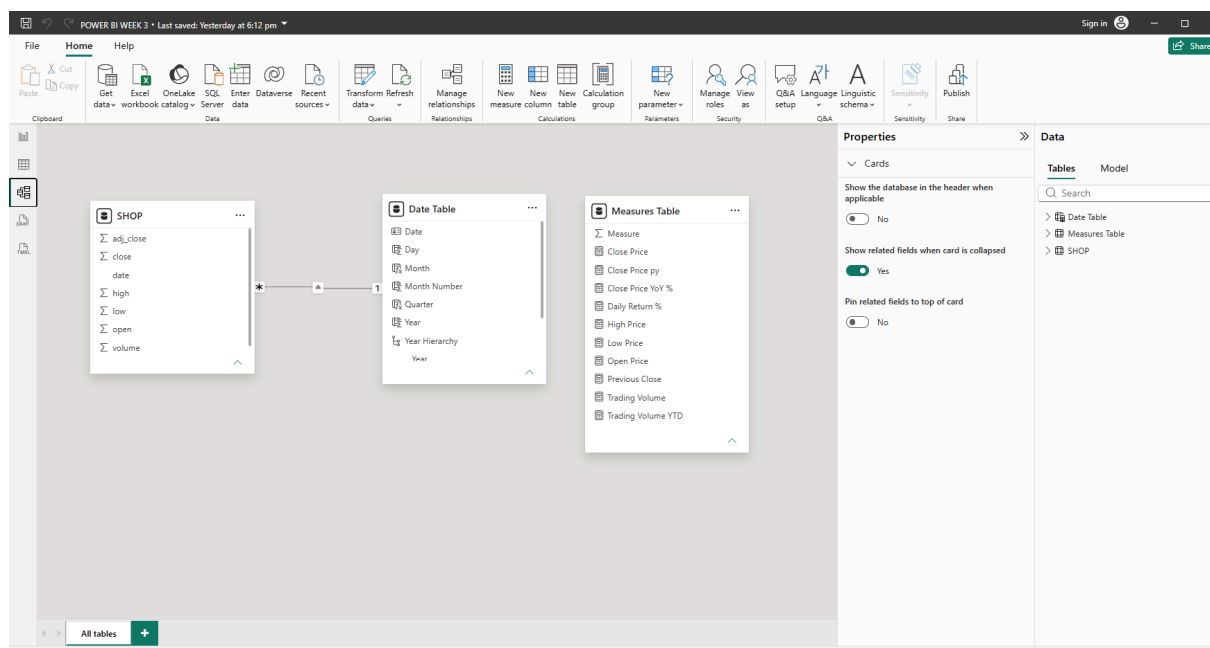
The **SHOP** table was connected to the **Date Table** using the Date field.

The relationship was established as:

Date Table[Date] → SHOP[date]

The relationship is a **one-to-many active relationship**, allowing the Date Table to be used for time-based filtering and analysis.

Date relationship between SHOP and Date Table.



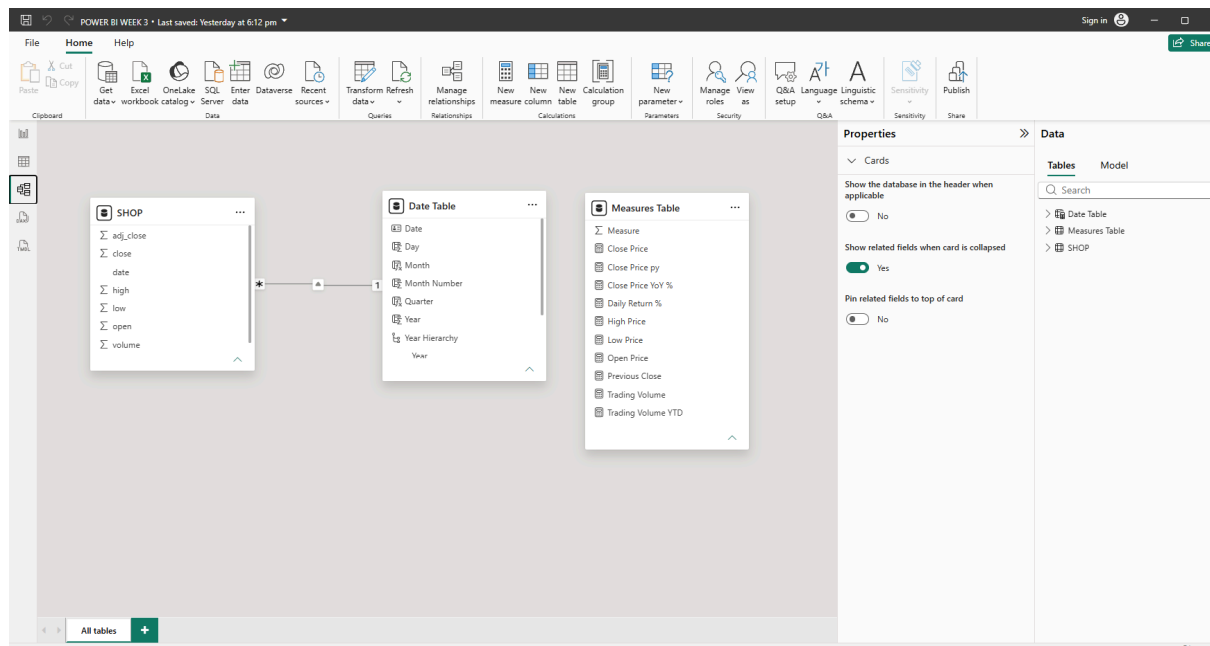
STEP 4 – CREATE THE MEASURES TABLE

A dedicated **Measures Table** was created to store the financial DAX measures separately from the stock data.

This helps organize the measures used throughout the Power BI report.

Creating the Measures Table.

Completed Measures Table in the model.



FINANCIAL DAX MEASURES

STEP 5 – CREATE OPEN PRICE MEASURE

The **Open Price** measure was created to calculate the average opening price of the stock.

Open Price =AVERAGE('SHOP'[open])

This measure is used for stock price analysis and KPI visualization.

STEP 6 – CREATE CLOSE PRICE MEASURE

The **Close Price** measure was created to calculate the average closing price.

Close Price =AVERAGE('SHOP'[close])

It is used to analyze and compare closing price levels over time.

STEP 7 – CREATE HIGH PRICE MEASURE

The **High Price** measure was created to calculate the average high price.

High Price =AVERAGE('SHOP'[high])

This measure supports yearly and time-based stock price comparison.

STEP 8 – CREATE LOW PRICE MEASURE

The **Low Price** measure was created to calculate the average low price.

Low Price =AVERAGE('SHOP'[low])

It is used together with High Price for stock price analysis.

STEP 9 – CREATE TRADING VOLUME MEASURE

The **Trading Volume** measure was created to calculate the total trading volume.

Trading Volume =SUM('SHOP'[volume])

This measure is used in KPI cards, charts, and the Decomposition Tree.

STEP 10 – CREATE PREVIOUS CLOSE AND DAILY RETURN

The **Previous Close** measure was created to obtain the previous-period closing value for return analysis.

The **Daily Return %** measure was also created to calculate the percentage change relative to the previous close.

These measures support stock return analysis.

STEP 11 – CREATE TIME-INTELLIGENCE MEASURES

Time-intelligence measures were created to perform time-based financial analysis.

Trading Volume YTD

Trading Volume YTD =TOTALYTD([Trading Volume],'Date Table'[Date])

The measure calculates the **Year-to-Date Trading Volume**.

Close Price PY

The **Close Price PY** measure was created to compare the closing price with the previous year.

Close Price YoY %

The **Close Price YoY %** measure was created to analyze the year-over-year percentage change in closing price.

The completed model includes:

- Trading Volume YTD
- Close Price PY
- Close Price YoY %
- Daily Return %

Financial measures and Trading Volume YTD.

STEP 12 – CREATE KPI CARDS

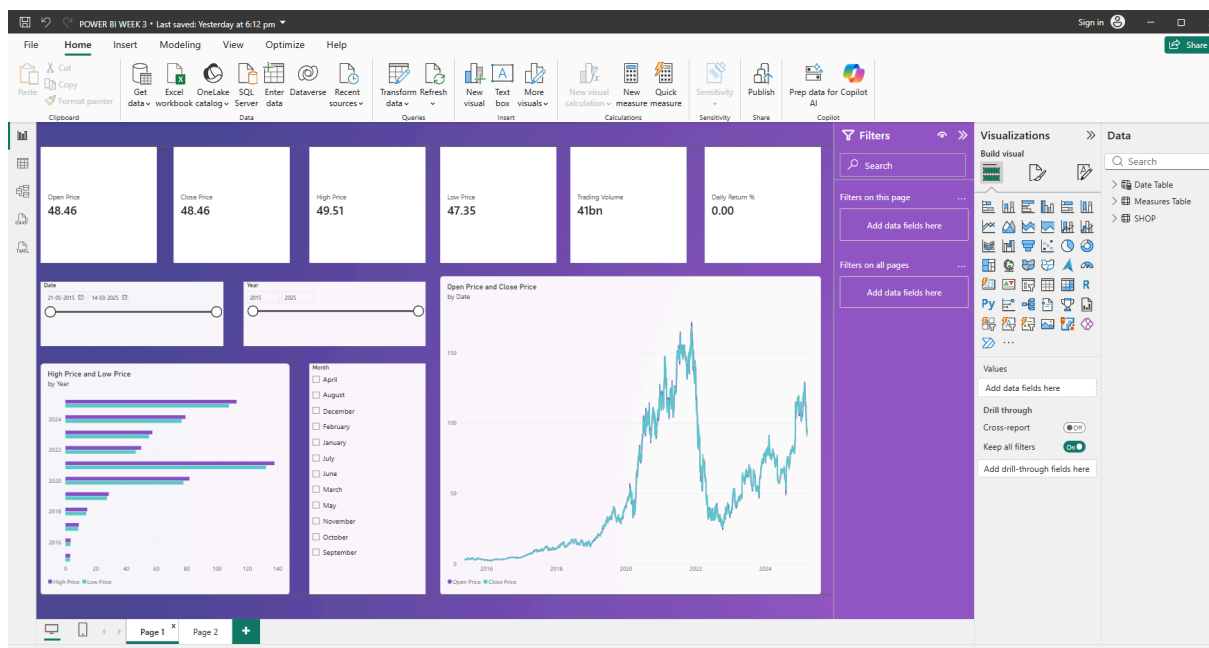
KPI/Card visuals were created to display the major stock indicators in a simple and understandable format.

The KPI cards display:

- Open Price
- Close Price
- High Price
- Low Price
- Trading Volume
- Daily Return %

These cards provide a quick overview of the stock market performance.

KPI cards showing stock indicators.

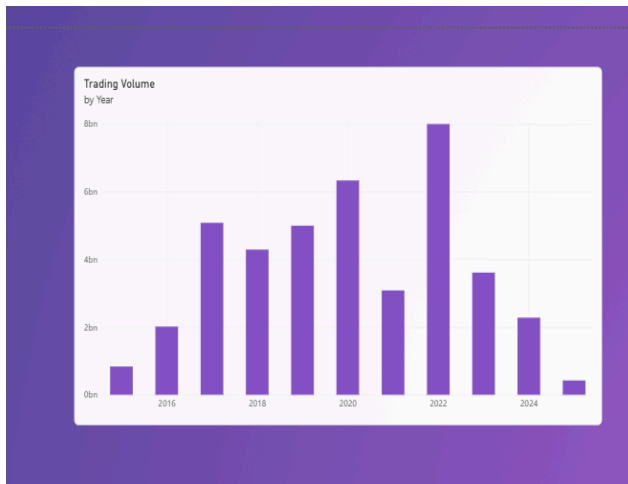


STEP 13 – CREATE YEARLY PRICE COMPARISON

A bar chart was created to compare stock price measures across different years.

The visual supports the comparison of stock price levels over time.

Year-based stock analysis page.



STEP 14 – CREATE DECOMPOSITION TREE

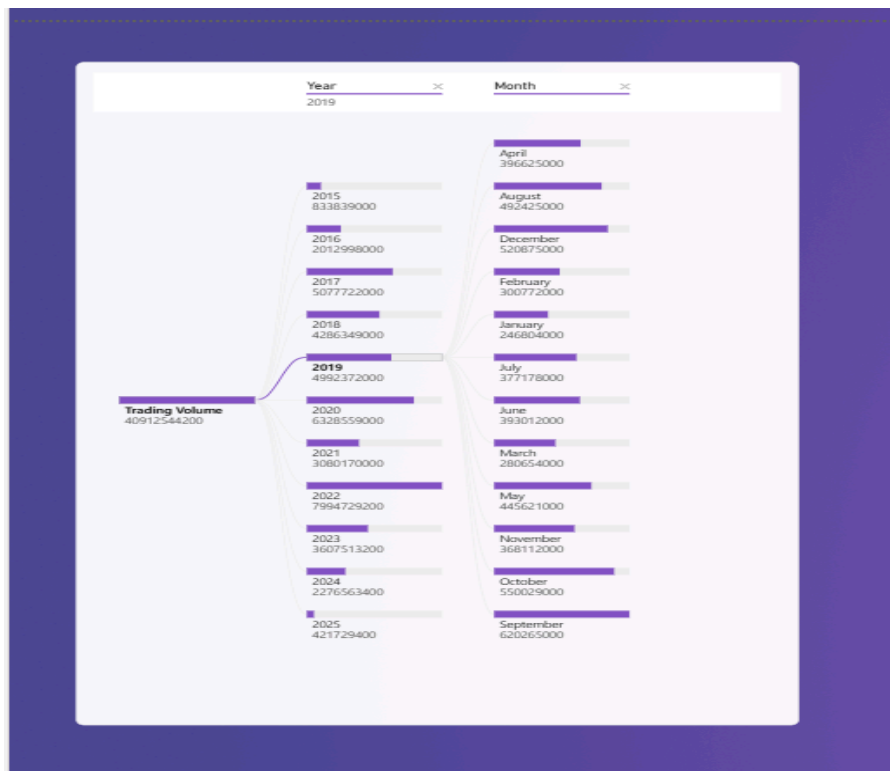
A **Decomposition Tree** was created to analyze Trading Volume interactively.

The visual was configured as:

- **Analyze:** Trading Volume
- **Explain by:** Year
- **Further breakdown:** Month

This allows trading volume to be explored from Year to Month.

Decomposition Tree with Year and Month analysis.



STEP 15 – CREATE DATE HIERARCHY

A custom date hierarchy was prepared for drill-down analysis.

The hierarchy follows:

Year → Quarter → Month → Day

This allows the user to move from yearly analysis to more detailed time periods.

Date hierarchy used for drill-down analysis.

STEP 16 – ADD SLICERS

Slicers were added to provide interactive filtering of the report.

The report supports date and year-based filtering, allowing users to select specific time periods for analysis.

Interactive report page with Year filtering.



STEP 17 – BUILD THE INTERACTIVE DASHBOARD

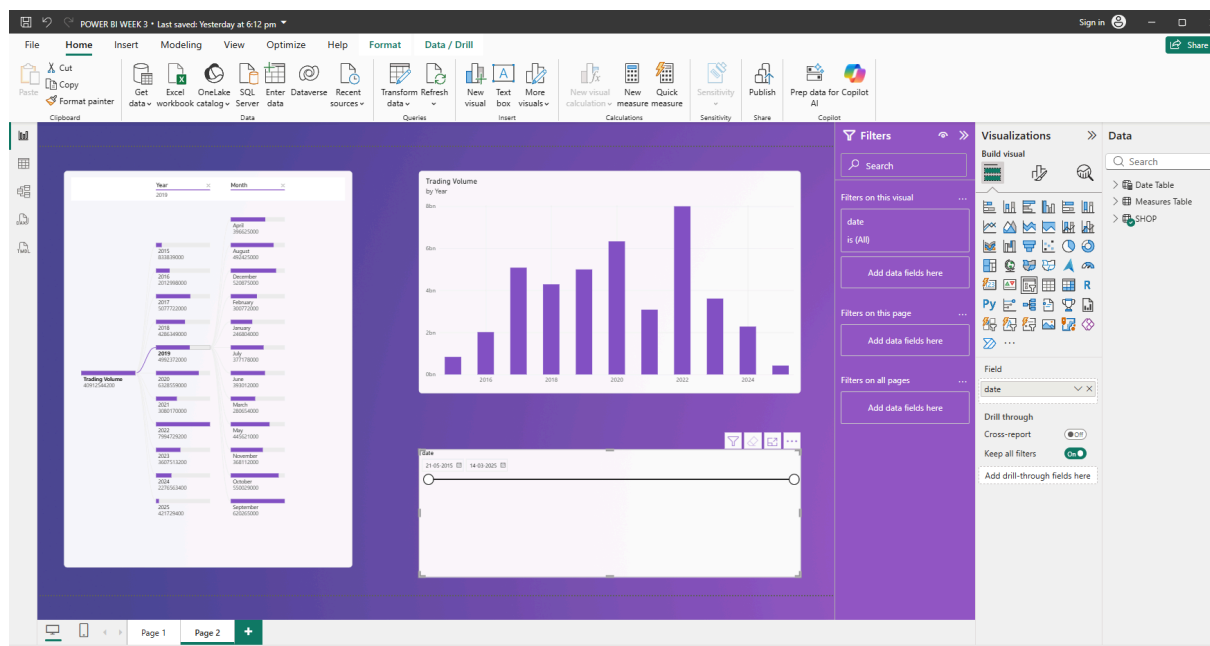
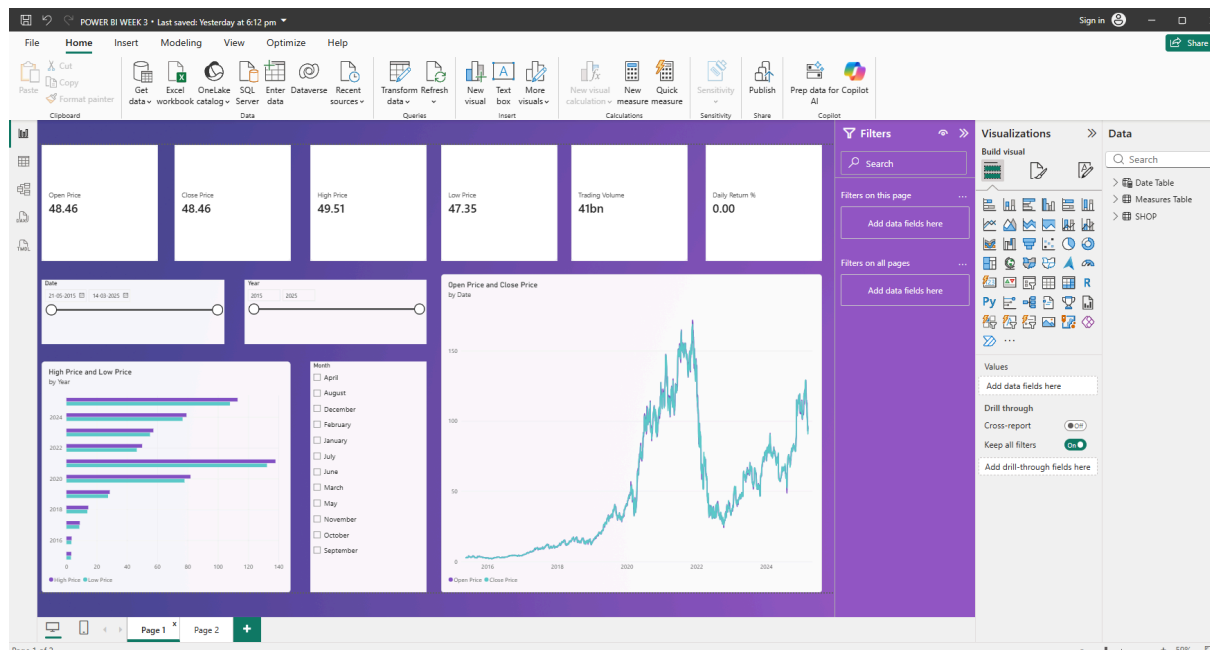
The final report combines multiple Power BI visuals into a structured financial analysis dashboard.

The dashboard includes:

- KPI Cards
- Year-based Price Chart
- Trading Volume Analysis
- Decomposition Tree
- Date/Year Filtering
- Date Hierarchy

These components allow the stock data to be explored interactively.

Final Shopify Stock Analysis report page.



CONCLUSION

The **Shopify Stock Data Analysis** project was successfully developed using Power BI for financial time-series analysis.

A dedicated **Date Table** and **Measures Table** were created, along with DAX measures for stock price, trading volume, return, and time-based analysis. Time-intelligence calculations such as **YTD**, **Previous Year**, and **Year-over-Year analysis** were also incorporated.

Interactive **KPI cards**, **charts**, **slicers**, **date hierarchy**, and **Decomposition Tree** visuals were used to create a structured financial analysis report.

The final dashboard provides an interactive way to explore **Open Price**, **High Price**, **Low Price**, **Close Price**, **Trading Volume**, and **return-related metrics** over different time periods.